

# TradeLens

Cross-Border Trade Compliance Platform

Tanzania – Zambia Corridor

---

## CBAP & PMP Project Document

Business Analysis & Project Management Plan

MVP Release Target: **30 June 2026**

---

|                         |                                          |
|-------------------------|------------------------------------------|
| <b>Business Analyst</b> | Given Kangwa<br>zda24b027@iitmz.ac.in    |
| <b>Project Manager</b>  | Patrick Vincent<br>zda24m007@iitmz.ac.in |
| <b>Institution</b>      | IIT Madras Zanzibar Campus               |
| <b>Version</b>          | 1.0 — May 2026                           |
| <b>Classification</b>   | Confidential                             |

*“Connecting Africa, one trade at a time.”*

Document Revision History

| Ver. | Date     | Author                         | Changes                            |
|------|----------|--------------------------------|------------------------------------|
| 1.0  | May 2026 | Given Kangwa & Patrick Vincent | Initial CBAP/PMP combined document |

Contents

|          |                                              |           |
|----------|----------------------------------------------|-----------|
| <b>I</b> | <b>CBAP — Business Analysis Document</b>     | <b>4</b>  |
| <b>1</b> | <b>Executive Summary</b>                     | <b>5</b>  |
| 1.1      | The Problem in Brief . . . . .               | 5         |
| 1.2      | Proposed Solution . . . . .                  | 5         |
| 1.3      | MVP Scope . . . . .                          | 6         |
| <b>2</b> | <b>Business Need &amp; Problem Statement</b> | <b>7</b>  |
| 2.1      | Context . . . . .                            | 7         |
| 2.2      | Root Cause Analysis . . . . .                | 7         |
| 2.3      | Formal Problem Statement . . . . .           | 7         |
| 2.4      | AS-IS vs TO-BE . . . . .                     | 8         |
| <b>3</b> | <b>Stakeholder Analysis</b>                  | <b>9</b>  |
| 3.1      | Stakeholder Register . . . . .               | 9         |
| 3.2      | Engagement Strategy . . . . .                | 10        |
| <b>4</b> | <b>Business Requirements</b>                 | <b>11</b> |
| 4.1      | Business Requirements Register . . . . .     | 11        |
| 4.2      | Functional Requirements . . . . .            | 12        |
| 4.2.1    | FR-01: User Management . . . . .             | 12        |
| 4.2.2    | FR-02: Certificate Registry . . . . .        | 12        |
| 4.2.3    | FR-03: Compliance Dashboard . . . . .        | 12        |
| 4.2.4    | FR-04: Customs Verification Portal . . . . . | 12        |
| 4.3      | Non-Functional Requirements . . . . .        | 12        |

|           |                                         |           |
|-----------|-----------------------------------------|-----------|
| <b>5</b>  | <b>Use Cases &amp; User Stories</b>     | <b>13</b> |
| 5.1       | Primary Use Cases . . . . .             | 13        |
| 5.2       | MVP User Story Backlog . . . . .        | 13        |
| 5.3       | Acceptance Criteria (Example) . . . . . | 14        |
| <b>6</b>  | <b>Solution Design Recommendations</b>  | <b>15</b> |
| 6.1       | Technology Stack . . . . .              | 15        |
| 6.2       | Key Data Entities . . . . .             | 15        |
| <b>II</b> | <b>PMP — Project Management Plan</b>    | <b>17</b> |
| <b>7</b>  | <b>Project Charter</b>                  | <b>18</b> |
| 7.1       | Project Information . . . . .           | 18        |
| 7.2       | Project Objectives . . . . .            | 18        |
| 7.3       | Scope Definition . . . . .              | 18        |
| 7.3.1     | In Scope . . . . .                      | 18        |
| 7.3.2     | Out of Scope . . . . .                  | 19        |
| 7.4       | Constraints and Assumptions . . . . .   | 19        |
| <b>8</b>  | <b>Work Breakdown Structure</b>         | <b>20</b> |
| 8.1       | Top-Level WBS . . . . .                 | 20        |
| 8.2       | Detailed WBS . . . . .                  | 20        |
| <b>9</b>  | <b>Project Schedule</b>                 | <b>23</b> |
| 9.1       | Sprint Plan . . . . .                   | 23        |
| 9.2       | Gantt Chart . . . . .                   | 24        |
| <b>10</b> | <b>Resource Plan</b>                    | <b>25</b> |
| 10.1      | Team Structure . . . . .                | 25        |
| 10.2      | Budget Estimate . . . . .               | 25        |
| <b>11</b> | <b>Risk Management Plan</b>             | <b>26</b> |
| 11.1      | Risk Register . . . . .                 | 26        |
| 11.2      | Critical Risk Response — R-02 . . . . . | 27        |
| <b>12</b> | <b>Quality Management Plan</b>          | <b>28</b> |
| 12.1      | Quality Standards . . . . .             | 28        |
| 12.2      | Testing Strategy . . . . .              | 28        |

|                                          |           |
|------------------------------------------|-----------|
| <b>13 Communication Plan</b>             | <b>29</b> |
| <b>14 Change Management Plan</b>         | <b>30</b> |
| <b>15 Closure &amp; Post-MVP Roadmap</b> | <b>31</b> |
| 15.1 MVP Closure Criteria . . . . .      | 31        |
| 15.2 Post-MVP Roadmap . . . . .          | 31        |
| 15.3 Investment Narrative . . . . .      | 31        |
| <b>A Glossary</b>                        | <b>33</b> |
| <b>B Document Sign-Off</b>               | <b>35</b> |

# Part I

## CBAP — Business Analysis Document

# Chapter 1

## Executive Summary

**TradeLens** is a web-first, mobile-supported digital platform that provides a unified compliance and certification registry for cross-border trade between Tanzania and Zambia, with a long-term vision to scale across EAC, SADC, and ECOWAS member states.

### 1.1 The Problem in Brief

Cross-border traders between Tanzania and Zambia face repeated manual compliance checks, paper-based certification, and fragmented regulatory systems. This causes increased clearance time (5–20 days), higher operational costs, and a chilling effect on SME participation in regional trade. There is no single, trusted digital system where a product's compliance status can be issued, stored, and verified across borders.

### 1.2 Proposed Solution

TradeLens solves this through three integrated modules:

1. **Unified Digital Certificate Registry** — single source of truth for product compliance certificates, recognised by customs on both sides of the corridor.
2. **Multi-Country Compliance Dashboard** — traders upload documents once; the system maps them against each destination country's requirements.
3. **Regulator & Customs Verification Portal** — border officers verify compliance status in real-time without re-inspecting paper documents.

## 1.3 MVP Scope

The MVP targets the Tanzania–Zambia corridor exclusively, focusing on agricultural and food products as the pilot product category.

# Chapter 2

## Business Need & Problem Statement

### 2.1 Context

The problem manifests during customs clearance in the Tanzania–Zambia corridor. Both countries are SADC members, yet their regulatory systems (TRA, ZRA, TBS, ZABS) are not integrated. Traders present physical certificates and repeat verification at each crossing.

### 2.2 Root Cause Analysis

#### Root Cause Summary (Fishbone):

- **People:** Low digital literacy among SMEs; officers reliant on manual processes.
- **Process:** No harmonised compliance framework; repeated verification per product.
- **Technology:** No interoperability between TRA, ZRA, TBS, and ZABS systems.
- **Policy:** Incomplete SADC trade protocol implementation.
- **Environment:** Limited connectivity at some border posts.

### 2.3 Formal Problem Statement

**The Problem of** fragmented manual compliance verification

**Affects** SME traders, exporters, manufacturers, and customs authorities in the Tanzania–Zambia corridor



**The Impact of Which Is** clearance times of 5–20 days, high operational costs, reduced SME market participation

**A Successful Solution Would** provide a single digital platform where certificates are issued once and verified instantly across both countries

## 2.4 AS-IS vs TO-BE

Table 2.1: Current State vs TradeLens Future State

| Dimension            | AS-IS                       | TO-BE (TradeLens)         |
|----------------------|-----------------------------|---------------------------|
| Certificate Issuance | Paper, per country          | Digital, issued once      |
| Verification         | Manual at each border       | Instant QR/ID lookup      |
| Document Submission  | Multiple times per shipment | Upload once, reuse        |
| Compliance Mapping   | Trader researches manually  | Auto-mapped by system     |
| Audit Trail          | Physical files, losable     | Immutable digital log     |
| Clearance Time       | 5–20 business days          | Target: 1–3 business days |

# Chapter 3

## Stakeholder Analysis

### 3.1 Stakeholder Register

| Stakeholder                  | Role            | Interest                      | Influence  |
|------------------------------|-----------------|-------------------------------|------------|
| SME Traders (TZ & ZM)        | Primary Users   | Faster clearance, lower costs | High       |
| Exporters/Manufacturers      | Primary Users   | Market expansion              | High       |
| Tanzania Revenue Authority   | Regulator       | Digital customs integration   | High       |
| Zambia Revenue Authority     | Regulator       | Digital customs integration   | High       |
| Tanzania Bureau of Standards | Standards Body  | Digital certificate issuance  | Medium     |
| Zambia Bureau of Standards   | Standards Body  | Digital certificate issuance  | Medium     |
| SADC Secretariat             | Policy          | Regional integration KPIs     | Medium     |
| Development Partners         | Funders         | Trade facilitation outcomes   | Medium     |
| Clearing Agents              | Secondary Users | Simplified documentation      | Low–Medium |
| TradeLens Dev Team           | Project         | MVP delivery                  | High       |

Table 3.1: Stakeholder Register

## 3.2 Engagement Strategy

Table 3.2: Stakeholder Engagement Matrix

| Stakeholder  | Strategy    | Key Action                         |
|--------------|-------------|------------------------------------|
| SME Traders  | Consult     | User interviews, usability testing |
| TRA / ZRA    | Collaborate | MOU for API data-sharing           |
| TBS / ZABS   | Collaborate | Certificate schema agreement       |
| SADC         | Inform      | Quarterly progress reports         |
| Dev Partners | Inform      | Pitch deck and demo                |

# Chapter 4

## Business Requirements

### 4.1 Business Requirements Register

| BR#   | Requirement                                                         | Priority    | Source      |
|-------|---------------------------------------------------------------------|-------------|-------------|
| BR-01 | Single digital certificate registry recognised by TRA and ZRA       | Must Have   | TRA, ZRA    |
| BR-02 | Traders upload documents once and reuse across both countries       | Must Have   | SME Traders |
| BR-03 | Customs officers verify compliance status in under 30 seconds       | Must Have   | TRA, ZRA    |
| BR-04 | System auto-maps documents against destination country requirements | Should Have | SME Traders |
| BR-05 | UI supports English, Swahili, and Nyanja                            | Should Have | SME Traders |
| BR-06 | All certificate transactions logged in immutable audit trail        | Must Have   | TRA, ZRA    |
| BR-07 | System generates compliance gap reports for traders                 | Could Have  | SME Traders |
| BR-08 | Mobile interface functional on 3G connectivity                      | Should Have | SME Traders |

Table 4.1: Business Requirements Register

## 4.2 Functional Requirements

### 4.2.1 FR-01: User Management

- Four roles: Trader, Standards Officer, Customs Officer, and System Admin.
- Registration requires national business registration number (BRELA / PACRA).
- MFA enforced for Customs and Standards roles.

### 4.2.2 FR-02: Certificate Registry

- TBS/ZABS officers issue digital certificates on the platform.
- Each certificate carries a unique QR code and SHA-256 hash fingerprint.
- Certificate validity periods enforced automatically.

### 4.2.3 FR-03: Compliance Dashboard

- Trader dashboard shows compliance status per destination country (green/amber/red).
- Missing documentation checklist with regulatory references displayed.
- Email/SMS notifications when certificates are approved or near expiry.

### 4.2.4 FR-04: Customs Verification Portal

- Officers search by trader ID, shipment number, or QR scan.
- Result returned within 30 seconds.
- Each verification logged with officer ID, timestamp, and decision.

## 4.3 Non-Functional Requirements

Table 4.2: Non-Functional Requirements

| Category       | Requirement                                                  |
|----------------|--------------------------------------------------------------|
| Performance    | API response $\leq$ 2 seconds (95th percentile)              |
| Availability   | 99.5% uptime SLA                                             |
| Security       | AES-256 at rest; TLS 1.3 in transit; OWASP Top-10 compliance |
| Scalability    | Horizontal scaling for 10,000 concurrent users               |
| Accessibility  | WCAG 2.1 AA                                                  |
| Connectivity   | Core workflows on 3G (progressive enhancement)               |
| Data Residency | Primary data stored in TZ/ZM or certified cloud regions      |

# Chapter 5

## Use Cases & User Stories

### 5.1 Primary Use Cases

**UC-1 Trader Onboarding & Document Upload** — Trader registers, uploads compliance documents, receives compliance status dashboard.

**UC-2 Certificate Issuance by Standards Body** — TBS/ZABS officer reviews documents and issues a digital certificate.

**UC-3 Customs Verification at Border** — Customs officer scans or searches a shipment and receives real-time compliance status.

### 5.2 MVP User Story Backlog

| US#   | As a ...          | I want to ...so that ...                                                                        |
|-------|-------------------|-------------------------------------------------------------------------------------------------|
| US-01 | SME Trader        | Register my business and upload product documents, so I can trade legally in the TZ-ZM corridor |
| US-02 | SME Trader        | See a compliance checklist per destination country, so I know what I need before shipping       |
| US-03 | SME Trader        | Receive an SMS when my certificate is approved, so I can plan shipment timing                   |
| US-04 | Standards Officer | Review and approve trader documents, so I can issue a digital certificate on the platform       |

| US#   | As a ...        | I want to ... so that ...                                                                                      |
|-------|-----------------|----------------------------------------------------------------------------------------------------------------|
| US-05 | Customs Officer | Scan a QR code and see compliance status instantly, so I can clear or hold a shipment without manual paperwork |
| US-06 | Customs Officer | View the full audit trail for a certificate, so I can confirm it has not been tampered with                    |
| US-07 | System Admin    | Manage user roles and permissions, so only authorised personnel can issue or verify certificates               |
| US-08 | Trader          | Download a PDF of my digital certificate to attach to physical shipment documents                              |

Table 5.1: MVP User Story Backlog

### 5.3 Acceptance Criteria (Example)

#### US-05 — Customs QR Verification

**Given** a customs officer scans or enters a valid certificate QR/ID,

**When** they submit the query,

**Then** the system returns a compliance status (Valid / Expired / Not Found) within 30 seconds, displaying trader name, product category, issuing body, expiry date, and a tamper-evident hash.

# Chapter 6

## Solution Design Recommendations

### 6.1 Technology Stack

Table 6.1: Recommended Technology Stack

| Layer          | Technology                    | Rationale                              |
|----------------|-------------------------------|----------------------------------------|
| Web Frontend   | React.js                      | Progressive web app; component reuse   |
| Mobile         | React Native                  | Code sharing with web; offline support |
| Backend API    | Node.js / FastAPI             | Lightweight; easy to containerise      |
| Database       | PostgreSQL                    | ACID compliance; strong JSON support   |
| Auth           | OAuth 2.0 + TOTP              | Role-based; MFA for regulatory roles   |
| Document Store | MinIO<br>(S3-compatible)      | Self-hosted; in-country data residency |
| Trust Layer    | SHA-256 hash +<br>timestamp   | Lightweight tamper-evidence            |
| Hosting (MVP)  | AWS af-south-1<br>(Cape Town) | Nearest compliant region to TZ/ZM      |
| Notifications  | Africa's Talking API          | SMS to TZ and ZM networks              |

### 6.2 Key Data Entities

- **User** — id, name, role, country, business\_reg\_no, email, phone
- **Product** — id, trader\_id, category, hs\_code, description
- **Document** — id, product\_id, doc\_type, file\_hash, uploaded\_at, status
- **Certificate** — id, document\_id, issuing\_body, issued\_at, expiry\_at, qr\_code,



hash

- **VerificationLog** — id, certificate\_id, officer\_id, verified\_at, outcome, border\_post

**Note on Integrations:** Full API integration with TRA and ZRA is a **post-MVP** activity requiring government MOUs. The MVP will use manual certificate issuance by enrolled TBS/ZABS officers as a proxy.

## Part II

### **PMP — Project Management Plan**

# Chapter 7

## Project Charter

### 7.1 Project Information

| Field             | Detail                                                                               |
|-------------------|--------------------------------------------------------------------------------------|
| Project Name      | TradeLens MVP                                                                        |
| Sponsor           | IIT Madras Zanzibar Campus                                                           |
| Project Manager   | Patrick Vincent ( <a href="mailto:zda24m007@iitmz.ac.in">zda24m007@iitmz.ac.in</a> ) |
| Business Analyst  | Given Kangwa ( <a href="mailto:zda24b027@iitmz.ac.in">zda24b027@iitmz.ac.in</a> )    |
| Start Date        | 1 May 2026                                                                           |
| Target Completion | 30 June 2026                                                                         |
| Estimated Budget  | \$4,000 USD (cloud, tooling, APIs)                                                   |
| Priority          | High                                                                                 |

### 7.2 Project Objectives

**OBJ-1** Deliver a functional web-based TradeLens MVP for the Tanzania–Zambia corridor by 30 June 2026.

**OBJ-2** Implement all Must-Have requirements (BR-01, BR-02, BR-03, BR-06).

**OBJ-3** Conduct a pilot with at least 5 SME traders and 2 regulatory officers.

**OBJ-4** Produce a post-MVP investment pitch deck supported by pilot metrics.

### 7.3 Scope Definition

#### 7.3.1 In Scope

- Web platform (desktop + mobile responsive)

- Trader, Standards Officer, and Customs Officer portals
- Digital certificate issuance and QR-based verification
- Tanzania–Zambia corridor and agricultural/food product category
- English language UI (Swahili/Nyanja as stretch goal)
- SMS notifications via Africa’s Talking

### 7.3.2 Out of Scope

- Direct API integration with TRA or ZRA legacy systems
- Native mobile app (iOS/Android)
- Payment processing
- Expansion beyond TZ-ZM corridor
- Blockchain implementation
- Additional product categories

## 7.4 Constraints and Assumptions

| Type       | Statement                                                             |
|------------|-----------------------------------------------------------------------|
| Constraint | MVP delivered by 30 June 2026                                         |
| Constraint | 2-person core team; external freelancers if required                  |
| Constraint | Maximum budget \$4,000 USD                                            |
| Assumption | At least 2 regulatory officers will participate in pilot              |
| Assumption | AWS af-south-1 meets data residency acceptability                     |
| Assumption | Both team members have development capability or access to developers |

# Chapter 8

## Work Breakdown Structure

### 8.1 Top-Level WBS

**WP-1 Project Management** — planning, monitoring, reporting

**WP-2 Business Analysis & Design** — requirements, wireframes, data model

**WP-3 Infrastructure Setup** — cloud, CI/CD, database

**WP-4 Development** — backend, frontend, integrations

**WP-5 Testing & Pilot** — QA, UAT, pilot

### 8.2 Detailed WBS

| WBS#                                        | Task                        | Owner   | Days    | Deliverable              |
|---------------------------------------------|-----------------------------|---------|---------|--------------------------|
| <b>WP-1: Project Management</b>             |                             |         |         |                          |
| 1.1                                         | Charter finalisation        | Patrick | 2       | Charter doc              |
| 1.2                                         | CBAP & PMP<br>documentation | Both    | 5       | This<br>document         |
| 1.3                                         | Weekly status reports       | Patrick | Ongoing | Status<br>reports        |
| <b>WP-2: Business Analysis &amp; Design</b> |                             |         |         |                          |
| 2.1                                         | Stakeholder interviews      | Given   | 5       | Interview<br>notes       |
| 2.2                                         | Requirements finalisation   | Given   | 3       | Requirements<br>register |

| WBS#                              | Task                            | Owner   | Days | Deliverable          |
|-----------------------------------|---------------------------------|---------|------|----------------------|
| 2.3                               | Wireframes (Figma)              | Given   | 5    | Wireframes           |
| 2.4                               | Data model design               | Both    | 3    | ER diagram           |
| 2.5                               | API contract design             | Patrick | 3    | OpenAPI spec         |
| <b>WP-3: Infrastructure Setup</b> |                                 |         |      |                      |
| 3.1                               | Cloud environment (AWS)         | Patrick | 2    | AWS setup            |
| 3.2                               | CI/CD pipeline (GitHub Actions) | Patrick | 2    | Pipeline             |
| 3.3                               | Database schema migration       | Patrick | 2    | PostgreSQL schema    |
| 3.4                               | Document storage (MinIO)        | Patrick | 1    | MinIO instance       |
| <b>WP-4: Development</b>          |                                 |         |      |                      |
| 4.1                               | Auth module (roles + MFA)       | Patrick | 4    | Auth service         |
| 4.2                               | Trader portal                   | Given   | 7    | Trader UI            |
| 4.3                               | Standards Officer portal        | Given   | 5    | Officer UI           |
| 4.4                               | Customs Verification portal     | Patrick | 5    | Customs UI           |
| 4.5                               | Certificate engine (QR + hash)  | Patrick | 5    | Certificate service  |
| 4.6                               | SMS notification integration    | Patrick | 2    | Notification service |
| 4.7                               | Compliance mapping engine       | Both    | 5    | Mapping module       |
| 4.8                               | Admin panel                     | Patrick | 3    | Admin UI             |
| <b>WP-5: Testing &amp; Pilot</b>  |                                 |         |      |                      |
| 5.1                               | Unit & integration testing      | Both    | 5    | Test reports         |
| 5.2                               | User Acceptance Testing         | Given   | 4    | UAT sign-off         |
| 5.3                               | Pilot (5 traders)               | Both    | 5    | Pilot feedback       |
| 5.4                               | Bug fixing & stabilisation      | Patrick | 3    | Release v1.0         |
| 5.5                               | MVP demo & handover             | Both    | 1    | Demo + pitch deck    |

| WBS# | Task | Owner | Days | Deliverable |
|------|------|-------|------|-------------|
|------|------|-------|------|-------------|

Table 8.1: TradeLens WBS

# Chapter 9

## Project Schedule

### 9.1 Sprint Plan

Table 9.1: Four-Sprint Plan — May to June 2026

| Sprint | Dates         | Goals                                               | Done When                               |
|--------|---------------|-----------------------------------------------------|-----------------------------------------|
| S1     | 1–14 May      | CBAP/PMP docs, wireframes, infra, auth              | Docs finalised; AWS live; auth working  |
| S2     | 15–28 May     | Trader portal, Standards portal, Certificate engine | Core portals functional; certs issuable |
| S3     | 29 May–11 Jun | Customs portal, compliance mapping, SMS, admin      | End-to-end user journey working         |
| S4     | 12–30 Jun     | Testing, UAT, pilot, fixes, demo                    | MVP stable; pilot done; pitch ready     |



9.2 Gantt Chart

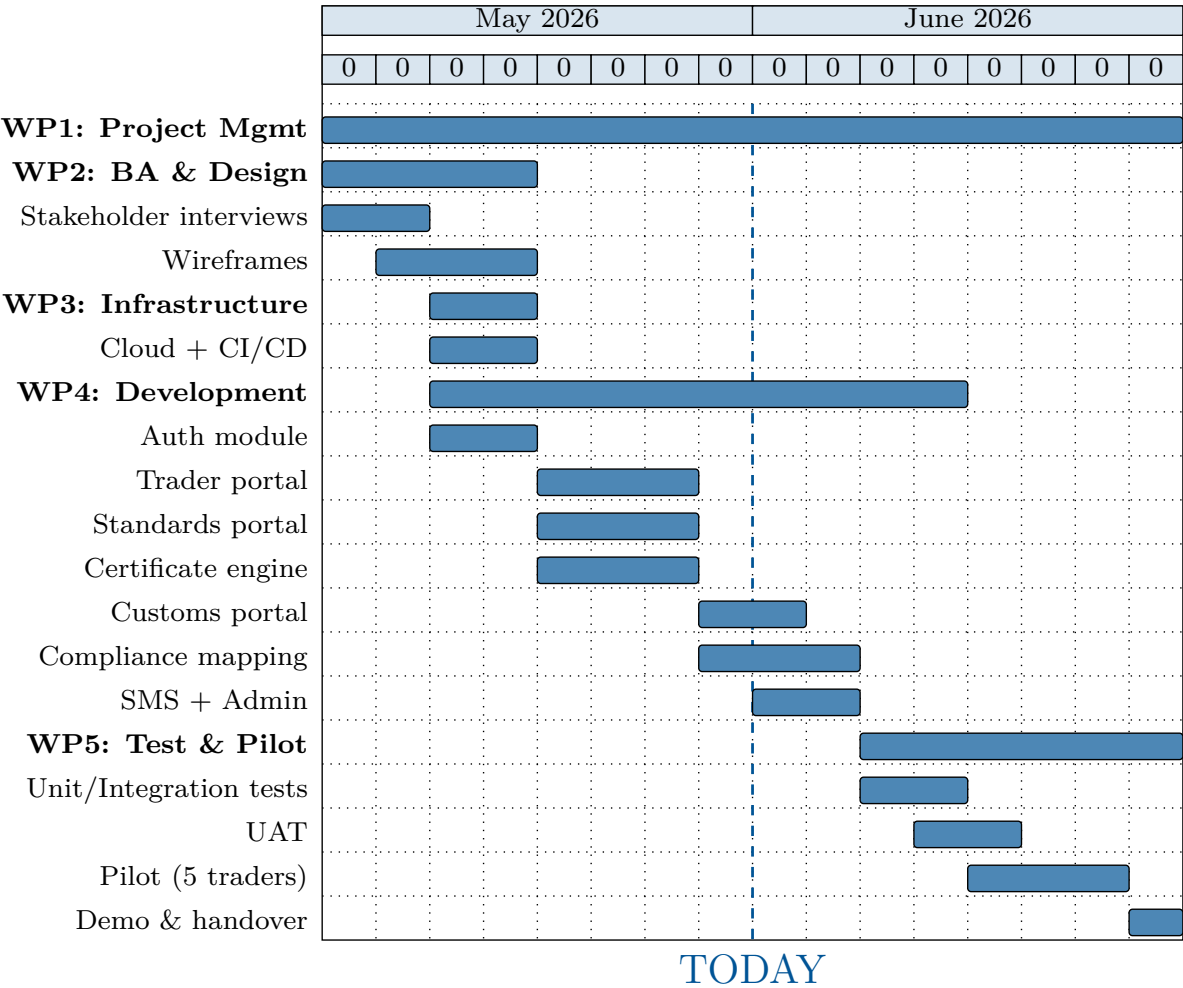


Figure 9.1: TradeLens MVP Gantt Chart — May to June 2026

# Chapter 10

## Resource Plan

### 10.1 Team Structure

Table 10.1: Project Team

| Name            | Role           | Responsibilities                                                                    |
|-----------------|----------------|-------------------------------------------------------------------------------------|
| Given Kangwa    | BA / UX Lead   | Requirements, wireframes, trader portal, UAT, pilot coordination                    |
| Patrick Vincent | PM / Tech Lead | Architecture, backend, DevOps, customs portal, certificate engine, project tracking |
| External (TBD)  | QA / Frontend  | Frontend support if needed; budget: \$500                                           |

### 10.2 Budget Estimate

Table 10.2: MVP Budget Breakdown

| Item                         | Cost (USD)     | Notes                    |
|------------------------------|----------------|--------------------------|
| AWS Cloud (2 months)         | \$800          | EC2, RDS, S3, CloudFront |
| Africa's Talking SMS API     | \$150          | ~3,000 SMS credits       |
| Domain and SSL               | \$50           | tradelens.io or similar  |
| Figma Pro (2 months)         | \$60           | UX design                |
| External QA/Frontend support | \$500          | Contingency only         |
| Miscellaneous tooling        | \$100          | GitHub, monitoring       |
| Contingency (20%)            | \$332          | Buffer                   |
| <b>Total</b>                 | <b>\$1,992</b> | Within \$4,000 cap       |

# Chapter 11

## Risk Management Plan

### 11.1 Risk Register

| R#   | Description                               | Prob.  | Impact | Rating   | Mitigation                                       |
|------|-------------------------------------------|--------|--------|----------|--------------------------------------------------|
| R-01 | Regulatory bodies do not engage for pilot | Medium | High   | High     | Use academic intermediary;<br>SADC contacts      |
| R-02 | Team capacity constraints                 | High   | High   | Critical | Prioritise<br>Must-Have; use<br>no-code tools    |
| R-03 | AWS latency to TZ/ZM border posts         | Low    | Medium | Low      | CDN caching;<br>early connectivity<br>test       |
| R-04 | Low SME digital literacy                  | Medium | Medium | Medium   | Simplified UX;<br>WhatsApp fallback              |
| R-05 | Budget overrun                            | Low    | Medium | Low      | 20% contingency;<br>scope reduction<br>protocol  |
| R-06 | Data security breach in pilot             | Low    | High   | Medium   | OWASP<br>compliance;<br>pen-test before<br>pilot |

| R#   | Description                       | Prob.  | Impact | Rating | Mitigation                                           |
|------|-----------------------------------|--------|--------|--------|------------------------------------------------------|
| R-07 | Schedule slippage<br>past 30 June | Medium | High   | High   | Weekly reviews;<br>drop Could-Have<br>features first |

Table 11.1: TradeLens Risk Register

## 11.2 Critical Risk Response — R-02

**Two-person team is the highest risk.** Mitigation steps:

- **Week 1:** Agree definitive scope; remove all Could-Have features from backlog.
- **Sprint 2:** If >3 days behind, activate external QA/Frontend budget (\$500).
- **Sprint 3:** If still behind, defer Swahili/Nyanja UI and compliance mapping engine to post-MVP.

# Chapter 12

## Quality Management Plan

### 12.1 Quality Standards

- API unit test coverage  $\geq 80\%$ .
- All user flows pass UAT sign-off by a pilot trader and a pilot officer.
- No P1 (Critical) or P2 (High) bugs in release build.
- Performance: API  $\leq 2s$  response at 100 concurrent users.
- OWASP Top-10 self-assessment completed before pilot.

### 12.2 Testing Strategy

Table 12.1: Testing Levels

| Level        | Approach                | Tools                       |
|--------------|-------------------------|-----------------------------|
| Unit Testing | Per module, automated   | Jest (JS), Pytest (Python)  |
| Integration  | API contract validation | Postman / Newman            |
| End-to-End   | Full user journey       | Playwright                  |
| UAT          | 3 traders + 2 officers  | Test script + feedback form |
| Performance  | 100 concurrent users    | k6                          |
| Security     | OWASP Top-10 checklist  | Manual + OWASP ZAP          |

# Chapter 13

## Communication Plan

Table 13.1: Communications Matrix

| Communication             | Audience              | Method              | Frequency          | Owner     |
|---------------------------|-----------------------|---------------------|--------------------|-----------|
| Sprint Review             | Both                  | Video call          | Bi-weekly          | Patrick   |
| Status Report             | Academic<br>Sponsor   | Email               | Weekly             | Patrick   |
| Stakeholder<br>Update     | TBS, ZABS             | Email               | Monthly            | Given     |
| Pilot Feedback<br>Session | Traders &<br>officers | In-<br>person/video | Once<br>(Sprint 4) | Given     |
| Task Tracking             | Internal              | GitHub<br>Projects  | Daily              | Both      |
| Incident Alerts           | Both                  | Slack/WhatsApp      | As needed          | Automated |

# Chapter 14

## Change Management Plan

**Step 1:** All change requests (CRs) logged in GitHub Issues with label `change-request`.

**Step 2:** Given (BA) assesses impact on requirements and scope.

**Step 3:** Patrick (PM) assesses impact on schedule and budget.

**Step 4:** Both team members approve or reject. If impact  $> 3$  person-days, escalate to sponsor.

**Step 5:** Approved CRs added to next sprint backlog; rejected CRs documented with rationale.

**Scope Freeze Date: 28 May 2026 (end of Sprint 2).**

No new Must-Have requirements after this date. All additions automatically go to post-MVP backlog.

# Chapter 15

## Closure & Post-MVP Roadmap

### 15.1 MVP Closure Criteria

- All Must-Have requirements (BR-01, BR-02, BR-03, BR-06) implemented and tested.
- UAT sign-off from at least 3 pilot traders and 1 pilot officer.
- Live demo conducted.
- Post-pilot metrics report produced.

### 15.2 Post-MVP Roadmap

Table 15.1: Post-MVP Roadmap

| Phase   | Timeline | Features                                                                   |
|---------|----------|----------------------------------------------------------------------------|
| Phase 2 | Q3 2026  | Swahili/Nyanja UI; WhatsApp integration;                                   |
| Phase 3 | Q4 2026  | TRA/ZRA API pilot; 2nd product category                                    |
| Phase 4 | 2026     | Full SADC expansion (Malawi, Mozambique); blockchain certificate anchoring |
|         |          | EAC and ECOWAS; payment integration; AI-powered compliance gap analysis    |

### 15.3 Investment Narrative

After MVP delivery, TradeLens will seek seed funding from the African Development Bank Innovation Fund, World Bank IFC, or impact investors, using pilot data as evidence. The pitch will centre on clearance time reduction, cost savings per shipment, willingness-to-pay data, and an addressable market of over 500,000 registered cross-border traders across



SADC.

# Appendix A

## Glossary

| Term    | Definition                                                                          |
|---------|-------------------------------------------------------------------------------------|
| BRELA   | Business Registrations and Licensing Agency (Tanzania)                              |
| CBAP    | Certified Business Analysis Professional — business analysis documentation standard |
| COMESA  | Common Market for Eastern and Southern Africa                                       |
| EAC     | East African Community                                                              |
| ECOWAS  | Economic Community of West African States                                           |
| HS Code | Harmonized System Code — international product classification for customs           |
| MOU     | Memorandum of Understanding                                                         |
| MVP     | Minimum Viable Product                                                              |
| OWASP   | Open Web Application Security Project                                               |
| PACRA   | Patents and Companies Registration Agency (Zambia)                                  |
| PMP     | Project Management Professional — project management planning standard              |
| SADC    | Southern African Development Community                                              |
| TBS     | Tanzania Bureau of Standards                                                        |
| TRA     | Tanzania Revenue Authority                                                          |
| TZ-ZM   | Tanzania–Zambia corridor abbreviation                                               |
| UAT     | User Acceptance Testing                                                             |
| ZABS    | Zambia Bureau of Standards                                                          |
| ZRA     | Zambia Revenue Authority                                                            |

| Term | Definition |
|------|------------|
|------|------------|

Table A.1: Glossary

# Appendix B

## Document Sign-Off

| Role             | Name                | Signature / Date |
|------------------|---------------------|------------------|
| Business Analyst | Given Kangwa        |                  |
| Project Manager  | Patrick Vincent     |                  |
| Academic Sponsor | IIT Madras Zanzibar |                  |

*TradeLens — Connecting Africa, one trade at a time.*

Document v1.0 · May 2026 · Confidential